Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	25 September 2026
Team ID	23KD1A1507
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2. The solution is built on the ServiceNow Now Platform: requesters submit network requests in the Service Portal (`/sp`), Flow Designer automates approval routing, dynamic record creation, and email notifications, while custom tables (`u_database_table`) and field-level ACLs enforce secure network task fulfillment.

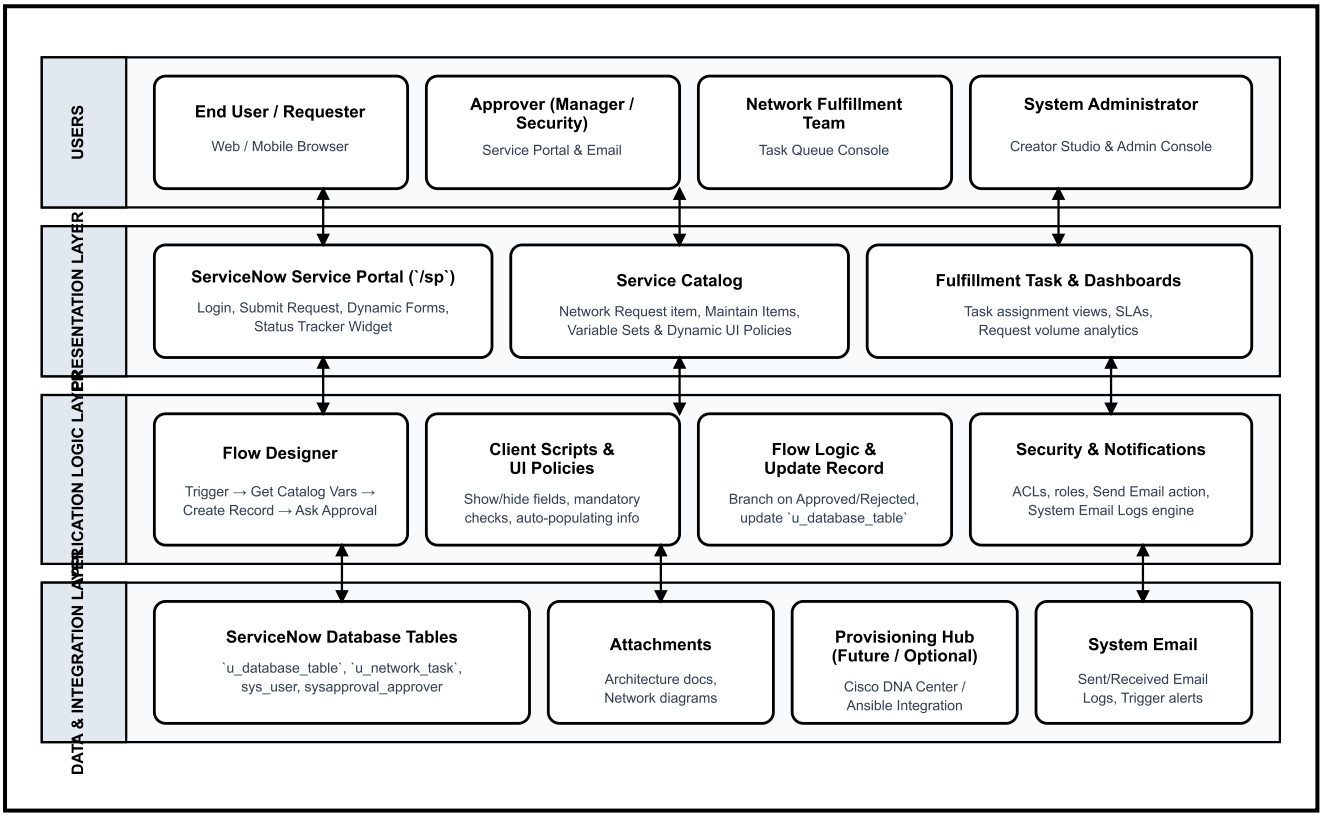


Figure 1: Technical architecture - Automated Network Request Management in ServiceNow

Table-1: Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web/mobile-browser interface through which requesters submit network requests, input variables, and view ticket status; staff interface for approvers and fulfillment teams.	ServiceNow Service Portal (`/sp`), Service Catalog, HTML, CSS, JavaScript, AngularJS widgets
2.	Application Logic-1	Business logic for dynamic form behavior (show/hide fields), mandatory field validations, variable set mappings, and auto-populating user data.	ServiceNow Client Scripts, UI Policies, Variable Sets, Server-side JavaScript
3.	Application Logic-2	Workflow automation of the request journey: Catalog Submission → Get Catalog Variables → Create Record → Ask for Approval → Update Record → Notifications.	ServiceNow Flow Designer / Workflow Editor
4.	Application Logic-3	Multi-level approval logic (Manager & Network Security approvals) and dynamic task generation for fulfillment engineers.	ServiceNow Flow Logic (If/Else), Approval Rules, Task Engine
5.	Database	Stores request records, variable data, approval states, and fulfillment tasks in custom tables with reference fields linking user and configuration data.	ServiceNow relational database (custom table `u_database_table`, `u_network_task`, `sys_user`, `sysapproval_approver`)
6.	Cloud Database	Managed relational database service on cloud with auto-numbering configuration, with no separate database installation required.	ServiceNow Cloud (Now Platform)
7.	File Storage	Storage of attached setup manuals, technical blueprints, and network architecture diagrams.	ServiceNow Attachments (`sys_attachment`)
8.	External API-1	Optional orchestration hook for auto-provisioning network ports and switch access via infrastructure automation tools.	IntegrationHub / REST Web Services (future: Cisco DNA Center / Ansible)
9.	External API-2	Delivery of automated booking confirmation emails, approval request alerts, and status update notifications to requesters and approvers.	ServiceNow Notification Engine (System Email Logs - Sent/Received)
10.	Machine Learning Model	Not used in current scope; request volume prediction and auto-resolution routing can be added in future phases.	Not applicable (future: ServiceNow Predictive Intelligence)
11.	Infrastructure (Server / Cloud)	Application deployment on cloud platform. Local Configuration: Not required (accessed through browser). Cloud Configuration: ServiceNow Personal Developer Instance (PDI) on the Now Platform.	ServiceNow Cloud instance (Now Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Modern responsive portal frameworks provided by the ServiceNow platform for building dynamic forms, catalog widgets, and status views.	AngularJS, Bootstrap (Service Portal), Creator Studio low-code components
2.	Security Implementations	Role-Based Access Control (RBAC) and field-level Access Control Rules (ACLs) to restrict read/write permissions on sensitive network fields; HTTPS/TLS for data in transit; full audit history logging for all requests and approval state changes.	ServiceNow Roles & ACLs, HTTPS/TLS, System Audit Logs, OWASP guidelines
3.	Scalable Architecture	Multi-tier architecture (presentation, application logic, data) on cloud platform; request categories, connection types, and approval chains are configuration-driven so expanding services requires no code changes.	ServiceNow Now Platform (3-tier architecture)
4.	Availability	Highly available cloud hosting with data-center redundancy managed by platform; submitted requests and approval logs remain persistently accessible to authorized users.	ServiceNow Cloud (high availability)
5.	Performance	Database indexing on key request fields (`u_database_table`), asynchronous email notifications, and Flow Designer execution engines optimized for rapid ticket creation and approval routing.	Database indexing, Flow Designer execution engine, scheduled jobs

References:

https://developer.servicenow.com
https://docs.servicenow.com
https://c4model.com/